

Storage in the Cloud TASK – 3

- 1. Create a new S3 bucket named 'my-zomato-images' and upload at least 3 sample image files representing restaurant menu items.**

Ans :

- Open the AWS Management Console and sign in.
- Navigate to Amazon S3.
- Click Create bucket.
- Enter the bucket name: my-zomato-images.
- Select an AWS Region (e.g., Asia Pacific (Mumbai) – ap-south-1).
- Leave the remaining settings as default (or configure them as required).
- Click Create bucket.
- Open the newly created my-zomato-images bucket.
- Click Upload.
- Click Add files.
- Select at least three restaurant menu item images (e.g., pizza.jpg, burger.jpg, pasta.jpg).
- Click Upload to upload the files.
- Wait for the upload to complete successfully.
- Verify that all three image files are visible inside the my-zomato-images bucket.

- 2. Set up a lifecycle policy on your 'my-zomato-images' S3 bucket to automatically transition all objects older than 30 days to the Glacier storage class.

Hint: Use the S3 Management Console's 'Lifecycle rules' section to define the transition action.**

Ans :

- Sign in to the AWS Management Console.
- Open the Amazon S3 service.
- Open the my-zomato-images bucket.
- Go to the Management tab.
- Under Lifecycle rules, click Create lifecycle rule.
- Enter a rule name (e.g., Transition-to-Glacier).
- Choose Apply to all objects in the bucket.
- Acknowledge the warning about applying the rule to all objects (if prompted).
- Under Lifecycle rule actions, select Transition current versions of objects between storage classes.
- Set the transition to 30 days after object creation.
- Choose Glacier Flexible Retrieval (or Glacier Instant Retrieval, depending on your assignment requirements).
- Leave the remaining settings as default.

**3. Simulate a cost optimization scenario: calculate and compare the estimated monthly storage cost for keeping 5GB of images in S3 Standard versus S3 Glacier using AWS Pricing Calculator.

Hint: Use the AWS Pricing Calculator website and note down the difference in INR.**

Ans :

- Open the AWS Pricing Calculator.
- Click Create estimate.
- Add the Amazon S3 service to your estimate.
- Configure the first estimate:
- Storage Class: S3 Standard
- Storage Amount: 5 GB
- Region: Same region as your S3 bucket (e.g., Asia Pacific – Mumbai)

- Note the estimated monthly storage cost for S3 Standard.
 - Add another Amazon S3 estimate.
 - Configure the second estimate:
 - Storage Class: S3 Glacier Flexible Retrieval
 - Storage Amount: 5 GB
 - Region: Same as above
 - Note the estimated monthly storage cost for S3 Glacier Flexible Retrieval.
 - Compare the two monthly storage costs.
 - Calculate the cost difference using:
 - $\text{Difference (INR)} = \text{S3 Standard Cost} - \text{S3 Glacier Cost}$
- 4. Add a new rule to your S3 bucket's lifecycle policy so that any objects with the prefix 'offers/' (e.g., 'offers/summer2024.jpg') are deleted automatically after 60 days.**

Ans :

- Sign in to the AWS Management Console.
- Open the Amazon S3 service.
- Open the my-zomato-images bucket.
- Go to the Management tab.
- Under Lifecycle rules, click Create lifecycle rule.
- Enter a rule name (e.g., Delete-Offers-After-60-Days).
- Choose Limit the scope of this rule using one or more filters.
- Select Prefix as the filter type.
- Under Lifecycle rule actions, select Expire current versions of objects.
- Set the expiration period to 60 days after object creation.
- Leave the remaining settings as default.

- Click Create rule or Save changes.

5. Use ChatGPT to generate a sample JSON configuration for an S3 lifecycle policy that transitions files to Glacier after 15 days and deletes them after 180 days. Paste the generated JSON and briefly explain each section.

Ans :

```
{
  "Rules": [
    {
      "ID": "TransitionAndExpireRule",
      "Status": "Enabled",
      "Filter": {
        "Prefix": ""
      },
      "Transitions": [
        {
          "Days": 15,
          "StorageClass": "GLACIER"
        }
      ],
      "Expiration": {
        "Days": 180
      }
    }
  ]
}
```

}

- Explanation of Each Section
- Rules
 - Contains the list of lifecycle rules applied to the S3 bucket.
- ID
 - A unique name for the lifecycle rule.
 - Example: "TransitionAndExpireRule".
- Status
 - Specifies whether the rule is active.
 - "Enabled" means the rule is applied.
- Filter
 - Defines which objects the rule applies to.
 - "Prefix": "" means the rule applies to all objects in the bucket.
- Transitions
 - Specifies when objects should move to another storage class.
 - "Days": 15 means objects are transitioned 15 days after upload.
 - "StorageClass": "GLACIER" moves the objects to the S3 Glacier storage class.
- Expiration
 - Specifies when objects are permanently deleted.
 - "Days": 180 means objects are automatically deleted 180 days after upload.

